

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but **do not open this question paper until you are ready to begin.**

This paper contains **27 multiple-choice questions**. There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all of the questions. Each question is worth one mark.

For each question, choose the one option you consider correct. If you make a mistake, erase thoroughly and try again.

You must complete the paper within the time limit.

You can use the question paper for rough working. Dictionaries and calculators are **NOT permitted**.

Please wait until you are ready before turning this page.

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Biology

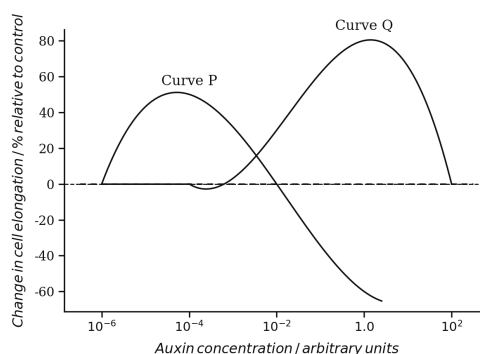
- 1 The graph shows the effect of auxin concentration on the elongation of cells in isolated root and shoot tissues of a plant species. One curve represents roots and the other represents shoots.

An intact seedling is placed horizontally in darkness. Due to gravity, auxin redistributes such that:

- the upper tissue layer in both the root and the shoot has an auxin concentration of 10^{-4} arbitrary units
- the lower tissue layer in both the root and the shoot has an auxin concentration of 1.0 arbitrary units

Which row in the table correctly identifies the curve representing roots and predicts the resulting direction of curvature for the root and the shoot?

	Curve representing roots	Direction of root curvature	Direction of shoot curvature
A	Curve P	downwards	upwards
B	Curve P	downwards	downwards
C	Curve P	upwards	upwards
D	Curve P	upwards	downwards
E	Curve Q	downwards	upwards
F	Curve Q	downwards	downwards
G	Curve Q	upwards	upwards
H	Curve Q	upwards	downwards



- 2** Warm water at 38°C is continuously discharged from an industrial cooling system into a cold, slow-flowing river.

Which of the following could occur in the river downstream of the discharge point?

- 1 An increase in the mass of dissolved oxygen held in a given volume of water at saturation.
- 2 An increase in the rate of aerobic cellular respiration per unit body mass in resident ectothermic organisms.
- 3 A reduction in the population size of fish species with high dissolved oxygen requirements.

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 3 The diagram shows a circular recombinant plasmid constructed by inserting a foreign gene of interest into a bacterial plasmid vector.

The original, non-recombinant plasmid contained three functional genes:

- gene *A*, coding for resistance to the antibiotic ampicillin
- gene *B*, coding for resistance to the antibiotic tetracycline
- gene *C*, coding for resistance to the antibiotic kanamycin

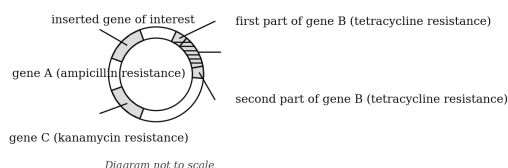
During the construction of the recombinant plasmid, the gene of interest was inserted into the coding region of gene *B*, as shown in the diagram.

Samples of bacteria containing only the recombinant plasmid and samples of bacteria containing only the original plasmid were separately inoculated onto four sets of agar plates:

- Plate 1: agar containing ampicillin and kanamycin
- Plate 2: agar containing ampicillin and tetracycline
- Plate 3: agar containing tetracycline and kanamycin
- Plate 4: nutrient agar containing no antibiotics

Which row in the table correctly identifies all the agar plates that will support the growth of each bacterial type?

	Agar plates supporting growth of bacteria containing only the recombinant plasmid	Agar plates supporting growth of bacteria containing only the original plasmid
A	1 and 4 only	1, 2, 3, and 4
B	1 and 4 only	1, 2, and 3 only
C	4 only	1, 2, 3, and 4
D	4 only	1, 2, and 3 only
E	1, 2, and 4 only	1, 2, 3, and 4
F	2 and 3 only	1, 2, 3, and 4
G	1 only	1, 2, and 3 only
H	1, 2, 3, and 4	1, 2, 3, and 4



- 4** Two strips of plant tissue, P and Q, are prepared for an experiment. The cells in Strip P have a mean internal solute concentration of 0.6 mol dm^{-3} . The cells in Strip Q have a mean internal solute concentration of 0.2 mol dm^{-3} .

Both strips are submerged in a beaker containing a sucrose solution with a concentration of 0.4 mol dm^{-3} .

Which option correctly predicts the initial change in turgor pressure for the cells in each strip?

- A** Strip P: Increases; Strip Q: Decreases
- B** Strip P: Increases; Strip Q: Increases
- C** Strip P: Decreases; Strip Q: Increases
- D** Strip P: Decreases; Strip Q: Decreases

- 5** Which of the following statements about human cells is/are correct?

1 An osteocyte contains the same nuclear genes as a mesenchymal stem cell from the same individual.

2 A mature spermatozoon contains half the number of chromosomes found in a secondary spermatocyte.

3 When an adult stem cell divides asymmetrically, it produces one stem cell and one daughter cell that can undergo differentiation.

(Assume that no mutations have occurred.)

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 6 In a plant, tall height is controlled by a dominant allele T and dwarf height by a recessive allele t . A pure-breeding tall plant is crossed with a pure-breeding dwarf plant. Two offspring from this cross are then crossed.

What proportion of their offspring is expected to be dwarf?

- A 0
B $\frac{1}{8}$
C $\frac{1}{4}$
D $\frac{1}{2}$
E $\frac{3}{4}$
F 1

- 7 The table shows the concentrations of substances in blood entering and leaving three different organs of a healthy adult who has been fasting for 24 hours:

- brain
- liver
- lungs

organ	glucose / mmol dm ⁻³ (entering)	oxygen / arbitrary units (entering)	urea / mmol dm ⁻³ (entering)	glucose / mmol dm ⁻³ (leaving)	oxygen / arbitrary units (leaving)	urea / mmol dm ⁻³ (leaving)
1	4.5	40	5.0	4.5	95	5.0
2	4.5	95	5.0	6.0	70	5.8
3	4.5	95	5.0	3.8	70	5.0

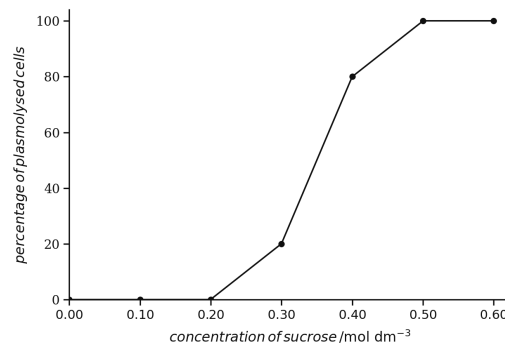
Which row in the table below correctly identifies the three organs?

	organ 1	organ 2	organ 3
A	lungs	brain	liver
B	lungs	liver	brain
C	liver	lungs	brain
D	liver	brain	lungs
E	brain	lungs	liver
F	brain	liver	lungs

- 8 Strips of plant epidermal tissue were placed into sucrose solutions of different concentrations. After reaching equilibrium, the percentage of plasmolysed cells was determined for each solution. Plasmolysis occurs when a plant cell loses water and its cytoplasm pulls away from the cell wall.

The graph shows the results obtained.

Which option correctly describes the effect of increasing the concentration of sucrose from 0.25 mol dm^{-3} to 0.40 mol dm^{-3} on the percentage of plasmolysed cells, and states the underlying cause?



- A increases by 70% due to the net movement of water out of the cells
- B increases by 70% due to the net movement of sucrose into the cells
- C increases by 80% due to the net movement of water out of the cells
- D increases by 80% due to the net movement of sucrose into the cells
- E decreases by 70% due to the net movement of water into the cells
- F decreases by 70% due to the net movement of sucrose out of the cells
- G decreases by 80% due to the net movement of water into the cells
- H decreases by 80% due to the net movement of sucrose out of the cells

- 9 The diagram shows part of the terrestrial nitrogen cycle involving four pools of nitrogen labeled **P**, **Q**, **R**, and **S**.

Which row correctly identifies boxes **P**, **Q**, **R**, and **S**?

	Atmospheric N_2	Ammonium ions in soil	Nitrate ions in soil	Organic nitrogen compounds in plants
A	P	R	S	Q
B	P	S	R	Q
C	Q	S	R	P
D	Q	R	S	P
E	S	R	Q	P
F	S	Q	R	P
G	Q	P	S	R
H	S	P	R	Q

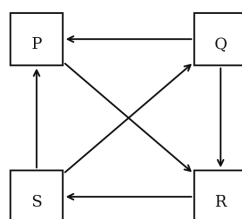


Diagram not to scale

- 10** Four distinct biological specimens (W, X, Y, and Z) were analysed. The results of the analysis are shown in the table below:

Sample	Genome type	Enclosed by a phospholipid bilayer	Contains ribosomes	Capable of synthesizing ATP
W	double-stranded DNA (circular)	yes	yes (70S)	yes
X	single-stranded RNA	yes	no	no
Y	double-stranded DNA (linear)	yes	yes (80S)	yes
Z	double-stranded DNA (linear)	no	no	no

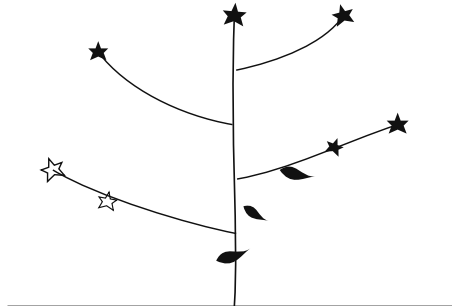
A student made the following statements about these specimens:

- 1 Sample X could be an enveloped virus, such as the influenza virus.
- 2 Sample W must be a mitochondrion and cannot be a free-living bacterium.
- 3 An antibiotic that selectively inhibits 70S ribosomes would directly inhibit protein synthesis in Sample W, but not in the cytoplasm of Sample Y.

Which of these statements is/are correct?

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 11** A seed is collected from a plant belonging to a true-breeding line that consistently produces red flowers. The seed is planted and grows into a new plant.
- This plant develops multiple branches, all of which bear red flowers, except for one single branch which exclusively bears white flowers, as shown in the diagram.



What is the most likely genetic explanation for the white-flowered branch?

- A** A somatic mutation occurred in a cell within the apical meristem of that specific branch.
- B** The parent plant was heterozygous, carrying a recessive allele for white flowers.
- C** The mutation for white flowers was present in the pollen grain that fertilised the parent flower.
- D** A localised deficiency of a specific mineral in the soil affected that branch.
- E** The mutation occurred in the zygote from which the entire plant grew.

- 12** In a species of mammal, coat pattern is determined by a single autosomal gene with two alleles, A^D and a .

- Embryos with the homozygous genotype $A^D A^D$ do not survive and die before birth.
- Heterozygous individuals ($A^D a$) have a striped coat.
- Homozygous individuals (aa) have a solid-colour coat.

Two striped individuals are mated and produce a large number of live offspring.

Which of the following statements is/are correct?

- 1 The probability that any individual live-born offspring has a striped coat is $\frac{2}{3}$.
- 2 The ratio of allele A^D to allele a among the live-born offspring is 1 : 2.
- 3 If a striped live-born offspring is mated with a solid-colour individual, the probability of an embryo dying before birth is $\frac{1}{4}$.

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

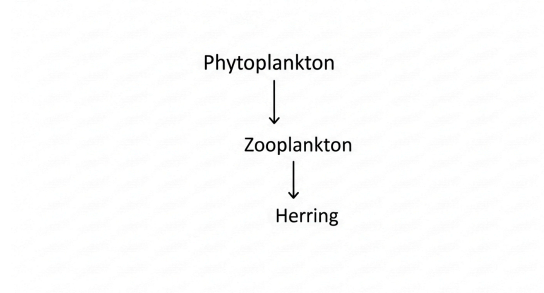
13 In honeybees (*Apis mellifera*), sex and fertility are determined by a single gene locus with multiple alleles (A_1, A_2, A_3 , etc.) via complementary sex determination:

- **Haploid** individuals develop from unfertilised eggs, possess only one allele at this locus (hemizygous), and develop into **fertile males**.
- **Diploid** individuals develop from fertilised eggs:
 - if they are **heterozygous** at this locus, they develop into **fertile females**;
 - if they are **homozygous** at this locus, they develop into **sterile males**.

Which row in the table correctly shows the sex and fertility of the five honeybees with the given ploidy and genotypes?

	Individual 1 (Haploid, A_1)	Individual 2 (Diploid, $A_1 A_2$)	Individual 3 (Diploid, $A_1 A_1$)	Individual 4 (Diploid, $A_2 A_3$)	Individual 5 (Diploid, $A_3 A_3$)
A	fertile male	fertile female	fertile female	fertile female	fertile female
B	fertile male	fertile female	fertile male	fertile female	fertile male
C	fertile male	fertile female	sterile female	fertile female	sterile female
D	fertile male	fertile female	sterile male	fertile female	fertile female
E	fertile male	fertile female	sterile male	fertile female	sterile male
F	fertile male	fertile female	sterile male	sterile female	sterile male
G	sterile male	fertile female	sterile male	fertile female	sterile male
H	sterile male	fertile female	fertile male	fertile female	fertile male

- 14** The diagram shows a simplified aquatic food web.



Which of the following statements about this food web is/are correct?

1. In this food web, the herring are primary consumers.
2. Approximately 90% of the energy from the zooplankton is transferred to the herring.
3. Decomposers make mineral ions from dead organic matter available to the phytoplankton.

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 15** The diagram shows a transverse section through a dicotyledonous plant leaf with four distinct regions labelled P, Q, R, and S.

Which of the following statements is/are correct?

- 1 The entire leaf structure shown is classified as an organ.
- 2 Region P and region Q can each be classified as a tissue.
- 3 The mature xylem vessel elements in region R contain mitochondria to provide energy for the active transport of water.

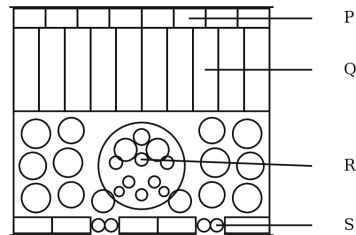


Diagram not to scale

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 16** Natural selection leads to evolutionary change in a population when a selection pressure acts upon heritable phenotypic variation, resulting in differential reproductive success.
- In which of the following scenarios will the environmental factor described act as a selection pressure that results in evolution by natural selection for the specified trait?
- 1 A population of clonal bacteria with identical genomes is exposed to an increased temperature within their survival range, where heat-shock protein production increases equally in all individuals via phenotypic plasticity.
 - 2 A population of field mice with heritable variation in fur colour is introduced to a light sandy habitat where predatory owls hunt by sight.
 - 3 A population of an annual plant in which variation in seed size is determined entirely by local soil depth with no genetic basis, growing in a region experiencing an increased frequency of summer droughts.
 - 4 A population of pathogenic bacteria containing a rare allele encoding a membrane efflux pump, exposed to therapeutic concentrations of an antibiotic that is expelled by this pump.
- A** 1 and 2 only
B 2 and 4 only
C 3 and 4 only
D 1, 2 and 4 only
E 2, 3 and 4 only
F 1, 2, 3 and 4
- 17** Researchers investigated the function of Protein Q in mitochondria. They compared wild-type yeast cells with a mutant strain lacking Protein Q.
- The table below shows the results of three measurements:
1. The rate of pyruvate oxidation (respiration) in intact mitochondria.
 2. The activity of Krebs cycle enzymes measured in lysed mitochondrial extracts.
 3. The percentage of soluble matrix proteins found in the cytoplasm (outside the mitochondria).
- | Measurement | Wild-type | Mutant (lacking Protein Q) |
|---|-----------|----------------------------|
| Rate of pyruvate oxidation (arbitrary units) | 100 | 5 |
| Activity of Krebs cycle enzymes in extracts (arbitrary units) | 100 | 98 |
| Matrix proteins found in cytoplasm (%) | < 1 | 65 |
- Which conclusion regarding the role of Protein Q is best supported by these data?
- A** Protein Q is an enzyme that catalyses a rate-limiting step in the Krebs cycle.
B Protein Q is a transcription factor required for the expression of genes encoding respiratory enzymes.
C Protein Q is essential for maintaining the structural integrity of the mitochondrial membranes.
D Protein Q acts as a specific carrier protein for the active transport of pyruvate into the matrix.
E Protein Q is an inhibitor that prevents the breakdown of Krebs cycle enzymes in the cytoplasm.

- 18** A diploid eukaryotic cell is homozygous for a gene encoding an enzyme.

The cell undergoes mitosis. During DNA replication prior to cell division, a single base substitution mutation occurs on one of the newly synthesised chromatids at this gene locus.

The cell successfully completes mitosis to produce two daughter cells: Cell 1 (which inherits the mutated chromatid) and Cell 2 (which inherits only non-mutated chromatids).

Assume no other mutations or chromosomal alterations occur.

Which of the following statements is/are correct?

- 1 The total number of gene loci in the nucleus is the same in both daughter cells.
- 2 Cell 1 has a greater number of different alleles for this gene than Cell 2.
- 3 Following DNA replication in the next cell cycle, the four chromatids carrying this gene in Cell 1 will all have identical base sequences.

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

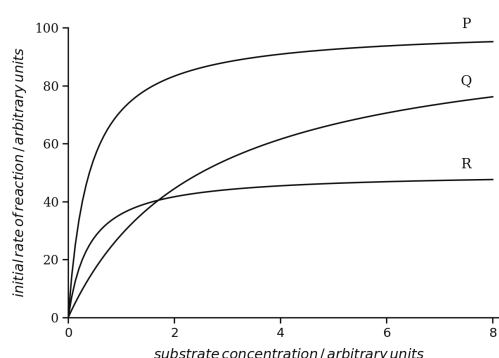
- 19** The graph shows the initial rate of an enzyme-catalysed reaction plotted against substrate concentration under three different experimental conditions: P, Q, and R.

In each experiment, the total concentration of enzyme present was identical.

- In experiment P, no inhibitor was added.
- In experiment Q, a constant concentration of substance X was added.
- In experiment R, a constant concentration of substance Y was added.

Which of the following statements is/are correct?

- 1 Substance X acts as a competitive inhibitor of the enzyme.
- 2 Increasing the substrate concentration can overcome the inhibitory effect of substance Y.
- 3 At a substrate concentration of 4 arbitrary units, a smaller proportion of enzyme molecules are bound to substrate in experiment Q than in experiment P.



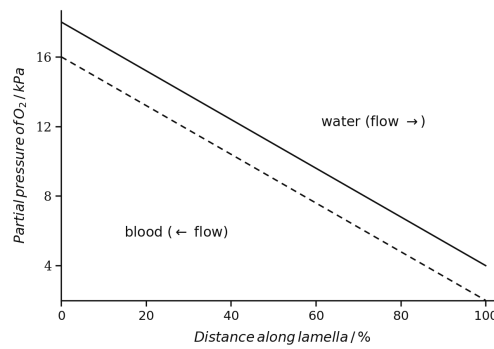
- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 20** The graph shows the partial pressure of oxygen ($p_{\text{ce}O_2}$) in water and in blood measured along the length of a secondary lamella in the gill of a bony fish.

Water flows across the lamella from distance 0 % to 100 %. Blood flows through the lamellar capillary in the opposite direction, from distance 100 % to 0 %.

Which of the following statements is/are correct?

- 1 Along the entire length of the lamella, net diffusion of oxygen occurs from the water into the blood.
- 2 Under the same flow conditions and entering partial pressures, if blood flowed in the same direction as water (concurrent flow), the maximum theoretical $p_{\text{ce}O_2}$ the blood could reach is 10.0 kPa.
- 3 The partial pressure gradient driving the diffusion of oxygen across the lamella is greatest at distance 0 %.



- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 21** A single-stranded RNA virus replicates inside host cells by first synthesising a complementary negative-sense RNA strand to form a double-stranded RNA intermediate.

The percentage of bases in the positive-sense single-stranded viral RNA genome was determined:

Base	Percentage in positive-sense RNA (%)
adenine (A)	32
uracil (U)	18
guanine (G)	28
cytosine (C)	*w*

Which of the following statements is/are correct?

- 1 In the complementary negative-sense RNA strand, 18% of the bases are adenine.
- 2 In the double-stranded RNA intermediate, 22% of the bases are cytosine.
- 3 In the double-stranded RNA intermediate, 50% of the bases are purines.

- A** 1 only
B 2 only
C 3 only
D 1 and 2 only
E 1 and 3 only
F 2 and 3 only
G 1, 2 and 3
H none of them

- 22** A double-stranded DNA molecule is 400 base pairs long and 32% of its bases are cytosine. One strand of this DNA molecule is transcribed to produce a single-stranded mRNA molecule that is 400 nucleotides long.

In this mRNA molecule:

- 22% of the bases are guanine
- 18% of the bases are uracil

Which of the following statements is/are correct?

- 1 The mRNA molecule contains 168 cytosine bases.
- 2 There are 1056 hydrogen bonds between the base pairs in the double-stranded DNA molecule.
- 3 Adenine makes up 36% of the bases in the mRNA molecule.

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 23** Three separate cultures of cells (Culture 1, Culture 2, and Culture 3) were established. At time $t = 0$ h, each culture contained 100 stem cells and no specialised (differentiated) cells.

All stem cells in each culture divided synchronously every 24 hours according to a specific mode of division:

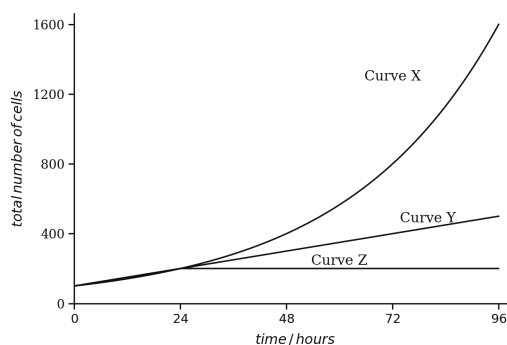
- **Culture 1:** each stem cell divides asymmetrically to produce one stem cell and one specialised cell.
- **Culture 2:** each stem cell divides symmetrically to produce two stem cells.
- **Culture 3:** each stem cell divides symmetrically to produce two specialised cells.

Specialised cells do not divide further, and no cells die during the 96-hour period.

The graph shows the total number of cells (stem cells plus specialised cells) over time for the three cultures, represented by curves X, Y, and Z.

Which row in the table correctly matches each culture to its corresponding curve on the graph, and gives the percentage of cells in Culture 1 that are stem cells at $t = 96$ h?

	Culture 1	Culture 2	Culture 3	Percentage of stem cells in Culture 1 at $t=96$ h
A	Curve Y	Curve X	Curve Z	20%
B	Curve Y	Curve X	Curve Z	25%
C	Curve Y	Curve Z	Curve X	20%
D	Curve Y	Curve Z	Curve X	50%
E	Curve X	Curve Y	Curve Z	20%
F	Curve X	Curve Y	Curve Z	25%
G	Curve Z	Curve X	Curve Y	20%
H	Curve Z	Curve Y	Curve X	50%

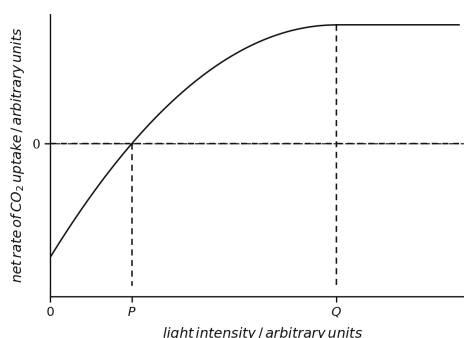


- 24** An experiment was carried out to investigate the effect of light intensity on the net rate of carbon dioxide (CO_2) uptake by a plant leaf at a constant temperature of 20°C and constant atmospheric CO_2 concentration. The graph shows the results obtained.

A student considered the graph and made the following conclusions:

- 1 Between light intensities 0 and P , no photosynthesis takes place.
- 2 At light intensity P , the rate of photosynthesis is equal to the rate of cellular respiration.
- 3 Between light intensities P and Q , light intensity is limiting the rate of photosynthesis.

Which of the student's conclusions is/are correct?



- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 25** A linear double-stranded DNA fragment contains 1500 base pairs and has a total of 3600 hydrogen bonds between its complementary nitrogenous bases.

Which of the following statements is/are correct?

- 1 Guanine accounts for 40% of the total nitrogenous bases in this DNA fragment.
- 2 There are 900 thymine bases in this DNA fragment.
- 3 There are 2998 phosphodiester bonds in the sugar–phosphate backbones of this DNA fragment.

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 26** Which of the following could be found inside a healthy pancreatic beta cell of an adult human?

- 1 circular DNA
- 2 gene for rhodopsin
- 3 cellulose

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 27** In a diploid organism, Process X produces cells required for growth and tissue repair, resulting in daughter cells that are genetically identical to the parent cell. Process Y produces cells required for sexual reproduction, resulting in daughter cells with half the number of chromosomes of the parent cell.

Which of the following statements correctly describes a difference in the chromosomal events between Process X and Process Y?

- A** DNA replication occurs prior to the start of Process X, but not prior to the start of Process Y.
- B** Sister chromatids separate and move to opposite poles in Process X, but not in Process Y.
- C** Homologous chromosomes pair up to form bivalents in Process Y, but not in Process X.
- D** Spindle fibres attach to the centromeres of chromosomes in Process Y, but not in Process X.
- E** Chromosomes condense and become visible in Process X, but not in Process Y.